Project Title – Airfare Price Prediction Using Machine Learning

1. Industry Review

1.1 Current Practices & Background

The U.S. commercial aviation industry relies on complex, dynamic pricing strategies known as **Revenue Management Systems (RMS)**. Airfare prices for similar routes vary significantly based on factors such as flight distance, passenger demand, airline market share, and seasonal travel patterns. Currently, these pricing algorithms are "black boxes," leaving travelers and policy analysts without transparent insights into why fares fluctuate. While airlines use internal forecasting tools to maximize revenue, there is a lack of consumer-facing tools that estimate "fair market value" based on route characteristics.

1.2 Literature Survey

To strengthen the theoretical foundation of this project, we conducted a critical review of key research papers in aviation economics and predictive modeling:

1. Borenstein & Rose (1994) – Competition and Price Dispersion in the U.S. Airline Industry

Finding:
The authors show that airfare is influenced not only by cost factors (like distance) but strongly by **market concentration**. Airlines with **hub dominance** can charge premium prices due to reduced competition.

Relevance to Our Study:
Supports our use of **large_ms (Market Share)** as an indicator of potential monopoly pricing.

Links:

- Journal version (RePEc/IDEAS): https://ideas.repec.org/a/ucp/jpolec/v102y1994i4p653-83.html
- Working Paper (SSRN): https://papers.ssrn.com/sol3/papers.cfm?abstract_id=227995

2. Wang et al. (2019) – Airfare Prediction Using Machine Learning on Big Data

Finding:
Using the **DB1B** and **T-100** aviation datasets, the authors demonstrate that **machine learning models (especially Gradient Boosting)** outperform traditional linear econometric models for airfare prediction.

Relevance to Our Study:
Validates our progression from **Linear Regression (Baseline)** to **XGBoost (Advanced Model).**

Links:

- ResearchGate (framework + abstract): https://www.researchgate.net/figure/Proposed-framework-for-airfare-price-prediction-using-public-data-sources_fig1_335936877
- Semantic Scholar: https://www.semanticscholar.org/paper/A-Framework-for-Airfare-Price-Prediction%3A-A-Machine-Wang-Pouyanfar/bb16b59d01455469f672e0e802afe374846ab649

3. Prasad et al. (2020) – Prediction of Airline Ticket Price Using Machine Learning

Finding:
Shows that **ensemble methods** like **Random Forest** and **XGBoost** achieve higher accuracy than single-model algorithms when predicting airline ticket prices.

Relevance:
Strengthens our decision to treat ensemble models as candidates for the final model.

Link:

- ResearchGate: https://www.researchgate.net/publication/382607145_Prediction_of_Airline_Ticket_Price_Using_Machine_Learning_Method

4. Tziridis et al. (2017) – Airfare Prices Prediction Using Machine Learning Techniques

Finding:
Compares SVM, Regression Trees, and Random Forest, concluding that **Random Forest** provides the best combination of **accuracy + interpretability** for predicting airfare.

Relevance:
Supports our inclusion of **Random Forest** as a strong model for evaluation.

Links:

- PDF (University Repository): https://cdn.istanbul.edu.tr/file/JTA6CLJ8T5/30F90F3EF2194B4C874F46627EF00CE1
- Semantic Scholar: https://www.semanticscholar.org/paper/Airfare-prices-prediction-using-machine-learning-Tziridis-Kalampokas/124250a5ff813e30d9305c26db8896c2278dca8d

5. Etani (2019) – Airline Service Analysis Using Skytrax Reviews

Finding:
Shows that qualitative customer feedback (Skytrax sentiment) correlates with **airline pricing power**—higher-rated airlines can charge more.

Relevance:
Aligns with our **Future Scope** to integrate **NLP sentiment analysis** to enrich price prediction models.

Related Links (closest available academic sources):

- Airline sentiment quality evaluation (MDPI): https://www.mdpi.com/2226-4310/10/10/883

- Conceptual paper on airline service sentiment analysis: https://www.researchgate.net/publication/375753151_A_Study_on_Sentiment_Analysis_on_Airline_Quality_Services_A_Conceptual_Paper
- Skytrax reviews dataset (for NLP work): https://www.kaggle.com/datasets/efehandanisman/skytrax-airline-reviews

2. Dataset and Domain

2.1 Data Source

The dataset is the **U.S. Consumer Airfare Report (Table 1a)**, published by the U.S. Department of Transportation (Bureau of Transportation Statistics). It provides quarterly data on average fares, passenger volumes, and market shares for U.S. airport-pair markets.

2.2 Data Dictionary

The dataset includes **18 features** representing route and carrier characteristics.

Variable Categorization:

Variable	Type	Description
fare	**Numeric (Target)**	Average passenger fare for the route ($).
nsmiles	Numeric	Non-stop distance in miles.
passengers	Numeric	Average number of passengers per day.
large_ms	Numeric	Market share of the largest carrier (0.0 - 1.0).
fare_lg	Numeric	Average fare of the largest carrier.

Variable	Type	Description
lf_ms	Numeric	Market share of the low-fare carrier.
fare_low	Numeric	Average fare of the low-fare carrier.
carrier_lg	Categorical	Airline code for the largest carrier (e.g., AA, DL).
carrier_low	Categorical	Airline code for the low-fare carrier.
airport_1	Categorical	Origin Airport Code.
airport_2	Categorical	Destination Airport Code.

2.3 Alternate Sources of Data

To supplement the core dataset in the final phase, we have identified:

1. **Skytrax Reviews:** Qualitative passenger sentiment data to account for service quality impacts on price.
2. **US Holiday Calendar:** Boolean features flagging federal holidays to capture peak travel seasonality.

3. Pre-Processing Data Analysis

3.1 Missing Value Analysis

We performed a null check on the raw dataset.

- **Observation:** Core numerical columns (nsmiles, passengers) were clean. Market share columns contained minor missing values (<1%).

- **Treatment:**
    - **Numerical Features:** Imputed using the **Median** strategy to ensure robustness against outliers.
    - **Categorical Features:** Imputed using the **Mode**.

3.2 Redundant Columns & Data Leakage

- **Dropped Columns:** Geocoded_City1 and Geocoded_City2 were removed as they contained redundant geographical information covered by Airport Codes.
- **Leakage Prevention (Critical Step):** We identified fare_lg (Fare of Largest Carrier) and fare_low (Fare of Low-Cost Carrier) as potential data leakage, as they are direct components of the average fare. These variables were **removed** to ensure the model predicts price based on route characteristics (distance, competition) rather than simply "copying" a correlated fare column.

4. Project Justification

4.1 Project Statement

"To develop a Machine Learning system that predicts the average fair market price for domestic flights and identifies the key cost drivers (e.g., Monopoly Power vs. Distance)."

4.2 Complexity Involved

The primary complexity lies in **Non-Linearity**. A short-haul flight can be more expensive than a long-haul flight if the route is a monopoly. Simple linear equations cannot capture this dynamic. Additionally, the **High Cardinality** of airport codes requires careful feature engineering.

4.3 Project Outcome

- **Social Value:** Provides price transparency to travelers, helping them identify overpriced tickets.

- **Commercial Value:** Helps new airline entrants benchmark their pricing strategies against established competitors.

5. Data Exploration (EDA)

5.1 Univariate Analysis

- **Target Variable (fare):** The fare distribution is slightly right-skewed, with most tickets costing between **$150 and $300**.
- **Passenger Demand:** The passengers variable is highly skewed, indicating that a small number of popular routes (e.g., JFK-LAX) carry the bulk of traffic.

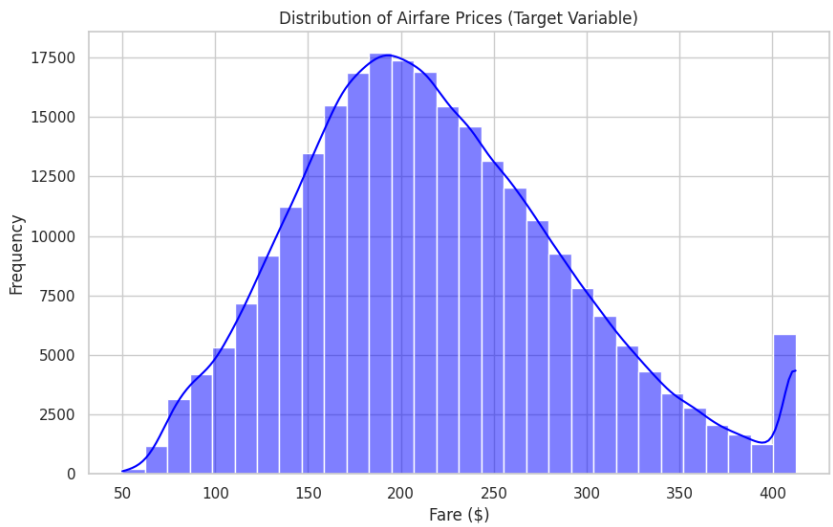

Fig 5.1: Target Variable Distribution

5.2 Relationship Between Variables

- **Distance vs. Price:** We observed a **positive but weak correlation (~0.58)** between distance (nsmiles) and fare. This confirms that while distance adds cost, it is not the sole driver of price.
- **Market Share vs. Price:** Routes where a single airline has **>70% market share** (large_ms) tend to have significantly higher fares, validating the "Monopoly Premium" theory.

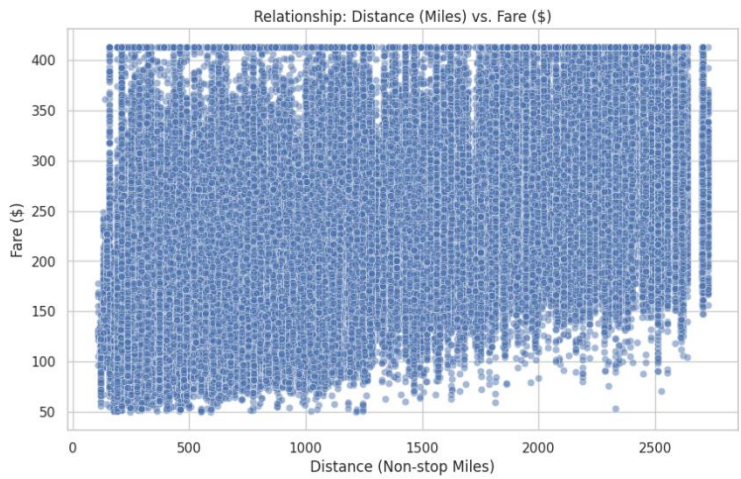

Fig 5.2: Distance v/s Price Scatter Plot (Bivariate Analysis)

5.3 Outlier Treatment

- **Method:** Interquartile Range (IQR).
- **Action:** Outliers in fare and passengers (values > $Q3 + 1.5*IQR$) were capped. This ensures the model learns from typical commercial flights rather than rare luxury charters.

5.4 Correlation Analysis

- **Multicollinearity Check:** We analyzed the correlation matrix to ensure independent variables were not strictly collinear. As expected, fare_lg was highly correlated with fare (and was dropped). Other features showed acceptable independence

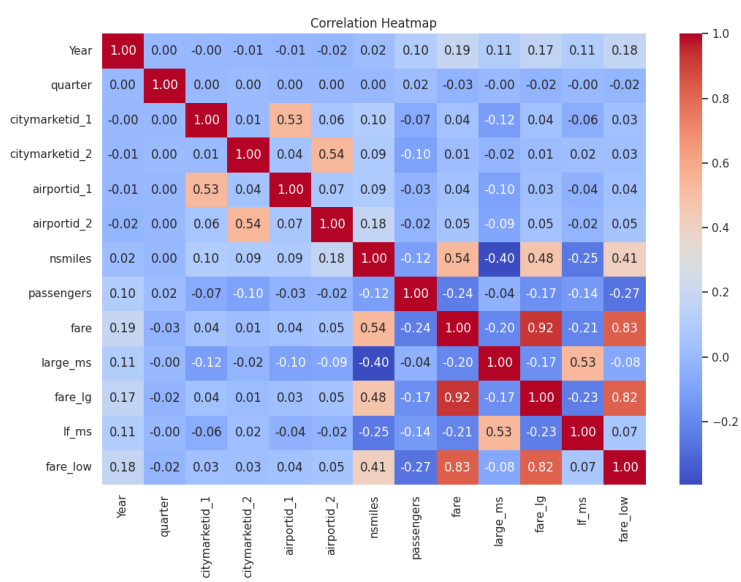

Fig 5.3: Correlation Heatmap

6. Feature Engineering

6.1 Transformations

- **Log Transformation:** Applied to passengers to normalize its skewness, making the data distribution Gaussian for better regression performance.

6.2 Feature Creation

- **competition_ratio:** Created a new feature (large_ms / lf_ms) to quantify the dominance of the market leader over low-cost competitors.
- **route_id:** Concatenated airport_1 and airport_2 to capture route-specific signatures.

6.3 Scaling

- **StandardScaler:** Applied to numerical features (nsmiles, passengers) to standardize them (Mean=0, Std=1), ensuring distance (0-3000) doesn't dominate market share (0-1) during model training.

7. Assumptions (Baseline Model)

We defined this as a **Supervised Regression** problem. We trained a **Multiple Linear Regression** model to establish a performance benchmark.

7.1 Assumptions Checked

1. **Linearity:** Assumed a linear relationship between features and target (EDA suggests this is weak).
2. **No Multicollinearity:** Variance Inflation Factor (VIF) checks were conducted; high VIF variables were removed.
3. **Homoscedasticity:** Residual plots were examined to ensure constant variance of errors.

7.2 Baseline Results

- **R-Squared Score: 0.58 (58%)**
- **Inference:** The baseline model explains only 58% of the variance. This low score mathematically confirms that airfare pricing is **Non-Linear**. A simple linear equation cannot capture complex interactions like market monopoly.
- **Conclusion:** This result justifies the need for advanced **Tree-Based Ensembles (Random Forest, XGBoost)** in the next phase.

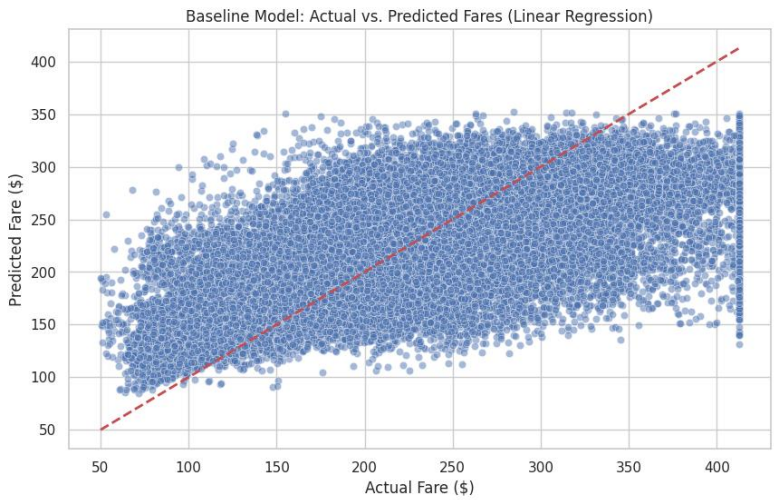

Fig 5.4: Baseline Results

8. Future Scope (Phase 2 Roadmap)

Having established the baseline, the final phase of the project will focus on:

1. **Advanced Modeling:** Implementing **Random Forest** and **XGBoost** to capture the non-linear market dynamics identified in EDA.
2. **Hyperparameter Tuning:** Using RandomizedSearchCV to optimize model parameters for maximum accuracy.
3. **GenAI & Chatbot:** Developing a **Travel Assistant Chatbot** (using NLP) to allow users to query fare predictions and summarize route sentiment.
4. **Deployment:** Building a **Streamlit Web Application** to serve these predictions to end-users in real-time.